EV



# LONGITUDINAL MODEL

Level 1 of the digital twin — built before the vehicle could be measured

C01 REVIEW

POL CRUAÑAS · 9 OCTOBER 2026

# Why a model before the data



**The coast-down could not run: both tyres punctured, and the logger's ESP32 was lost to a ground loop.**

C01 PLAN · SEPTEMBER

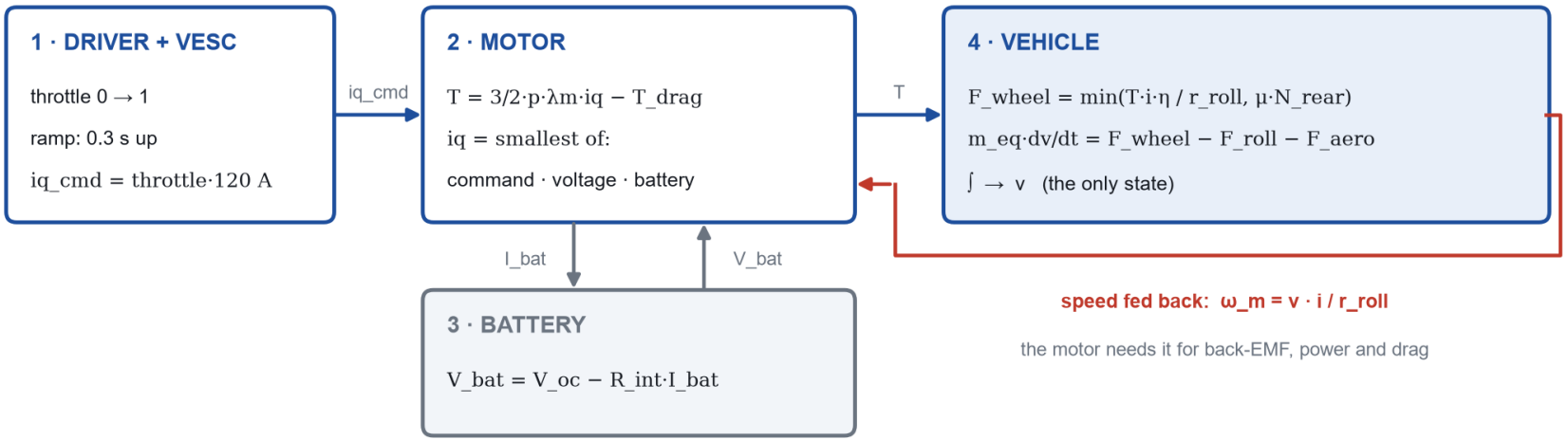
measure → fit → model

DONE · AFTER THE PIVOT

model → predict + freeze → measure → C02

**Written down first, so the test can prove it wrong.**

# The model: four blocks, one state



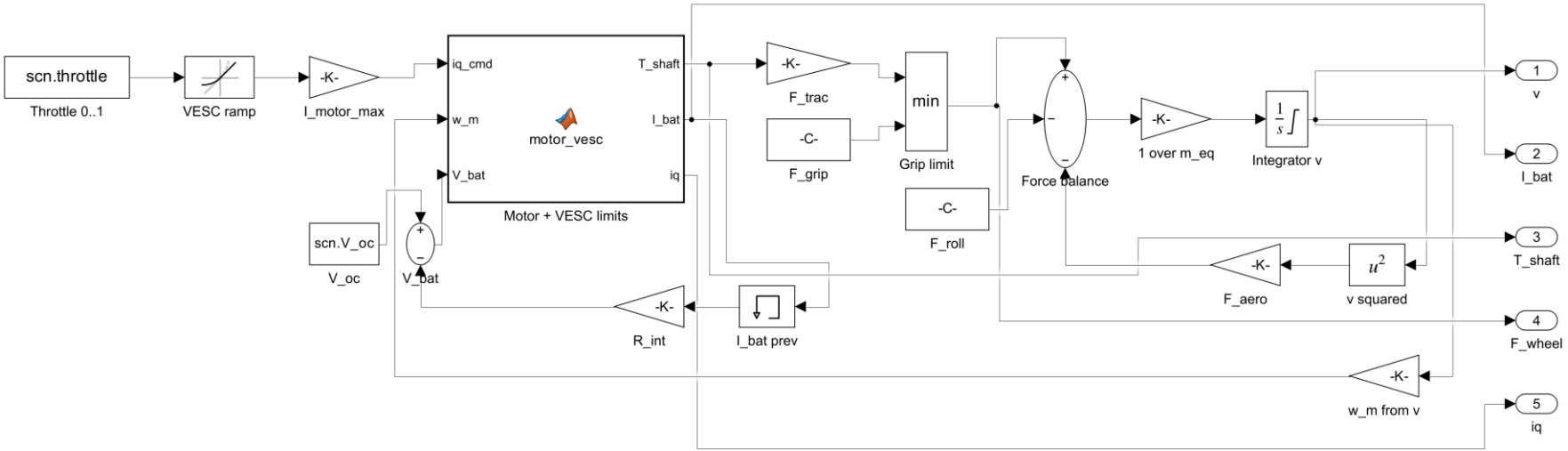

## SANITY CHECK · MOTOR ONLY, WHEELS IN THE AIR

**5448 rpm**
model, wheels in the air, 52.77 V

**5480 rpm**
measured, C00 no-load (16 440 ERPM ÷ 3)

**−0.6 %**
λm + voltage ceiling + drag — not the vehicle

# The same model in Simulink (L1)




Built by script (build_longitudinal_L1.m): every block and wire is text in Git. L2 = same physics in 4 subsystems, identical output.

# The force balance, with numbers

$$m_{eq} \cdot dv/dt = F_{wheel} - F_{roll} - F_{aero}$$

| TERM | LAW | VALUE | INPUTS |
|---|---|---|---|
| **F_wheel** | $T \cdot i \cdot \eta / r_{roll}$ | **≤ 463 N** | i = 8.36 · η = 0.95 · r = 0.155 m |
| **F_roll** | $Crr \cdot m \cdot g$ | **32 N** | Crr = 0.030 · m = 110 kg |
| **F_aero** | $½ \cdot \rho \cdot CdA \cdot v^2$ | **25 N at 30 km/h** | CdA = 0.60 m² · ρ = 1.20 kg/m³ |
| **grip** | $\mu \cdot N_{rear}$ | **421 N** | μ = 0.60 (dirt) · rear share 0.65 |
| **m_eq** | $m + J_{wheels}/r^2 + J_{rotor} \cdot i^2/r^2$ | **114.5 kg** | 110 + 4 wheels (ASSUMED) + 0.45 rotor (MEASURED) |

**At top speed, rolling and aero are about equal. Below it, rolling resistance dominates.**

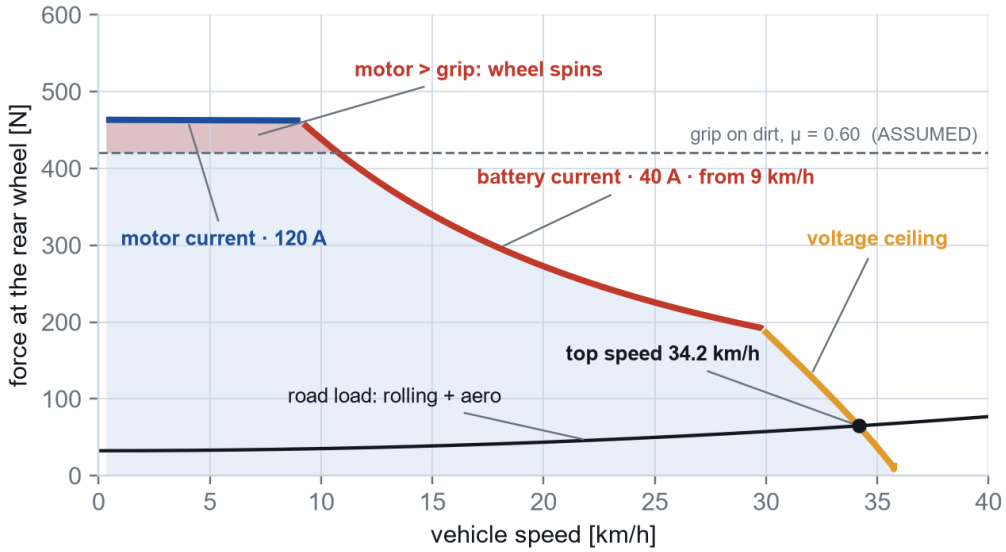
EV

# Where every number comes from

| MEASURED | | VESC ESTIMATE | | CONFIG | | ASSUMED | |
|---|---|---|---|---|---|---|---|
| pole pairs | | phase resistance | | motor current | | rolling radius | |
| **3** | Hall count | **28 mΩ** | detection, 3 runs | **120 A** | typed in VESC | **0.155 m** | 13 in, −6 % squash |
| flux linkage λm | | inductance | | battery current | | Crr | |
| **16.9 mWb** | scope, 3 methods | **106 µH** | detection, 3 runs | **40 A** | = BMS rating | **0.030** | soft knobby tyre |
| rotor drag | | | | max duty | | CdA | |
| **63 mN·m** | spin-down, + viscous | | | **0.95** | typed in VESC | **0.60 m²** | upright rider |
| masses | | | | throttle ramp | | grip µ | |
| **38 + 72 kg** | scale | | | **0.3 s** | typed in VESC | **0.60** | dirt, no tyre data |
| gear | | | | | | chain η | |
| **92/11** | teeth counted | | | | | **0.95** | roller chain |
| | | | | | | pack R_int | |
| | | | | | | **0.10 Ω** | no datasheet |

**Every ASSUMED value carries a range. The sensitivity sweep runs over exactly those ranges.**

# Where the limits are




**MOTOR CURRENT**

$iq \leq 120\ A$

T = 3/2·p·λm·iq = 9.1 N·m

**BATTERY CURRENT**

$I_bat \leq 40\ A$

≈ 1.8 kW, from 9 km/h

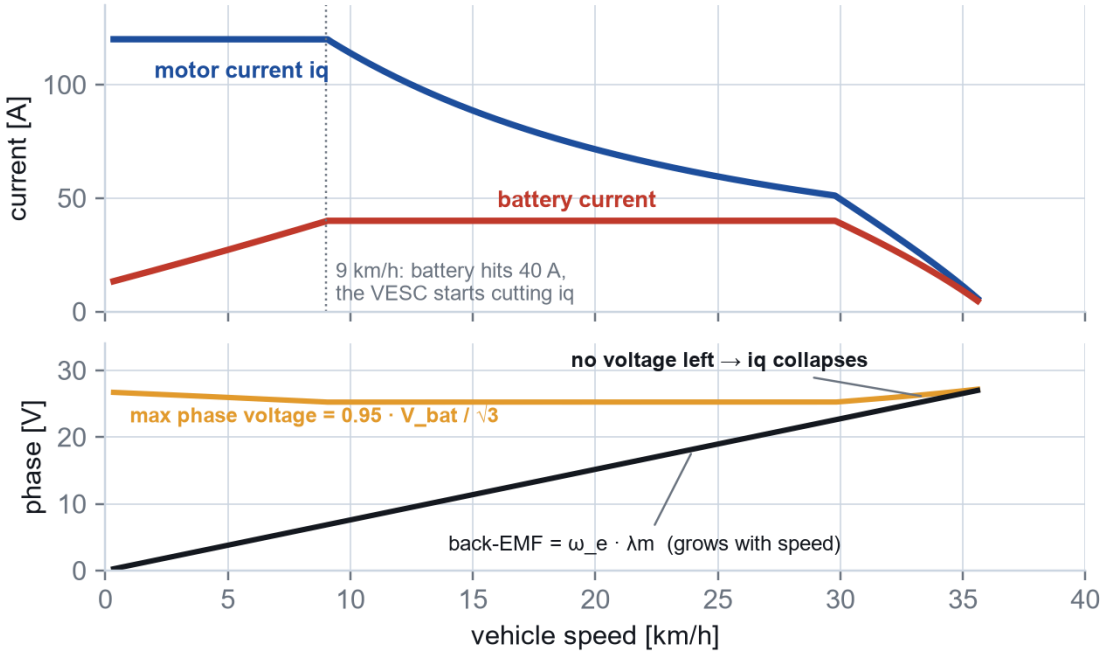
**VOLTAGE**

$|v| \leq 0.95 \cdot V_bat / \sqrt{3}$

C00 check: 28.9 V reached

**On dirt the tyre limits the launch (421 N < 463 N). From 9 km/h: the battery. At the top: voltage.**

# Motor current is not battery current




**POWER IS CONSERVED**

V_bat · I_bat ≈ 3/2 · v_q · iq. Low speed → low motor voltage → 120 A of motor is little battery current.

**FROM 9 KM/H**

The battery reaches 40 A. The VESC cuts iq so it stays there: constant power.

**NEAR THE TOP**

The back-EMF reaches the voltage the inverter can make. No voltage left to push current: iq collapses.

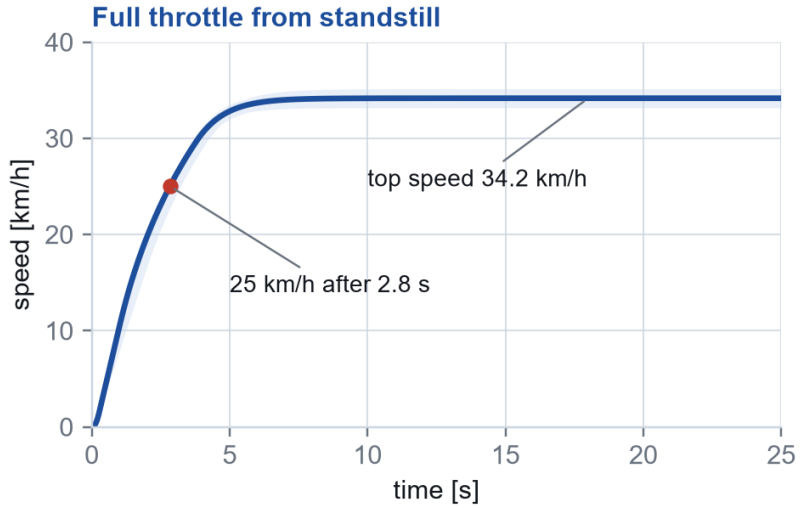
# Predictions, frozen before the test

**34.2 ± 1.2 km/h**
top speed, at 50 V

**2.8 ± 0.5 s**
0 → 25 km/h, full throttle

**17.2 ± 5.6 s**
coast-down 30 → 5 km/h

**9.0 ± 0.6 km/h**
where the 40 A limit starts

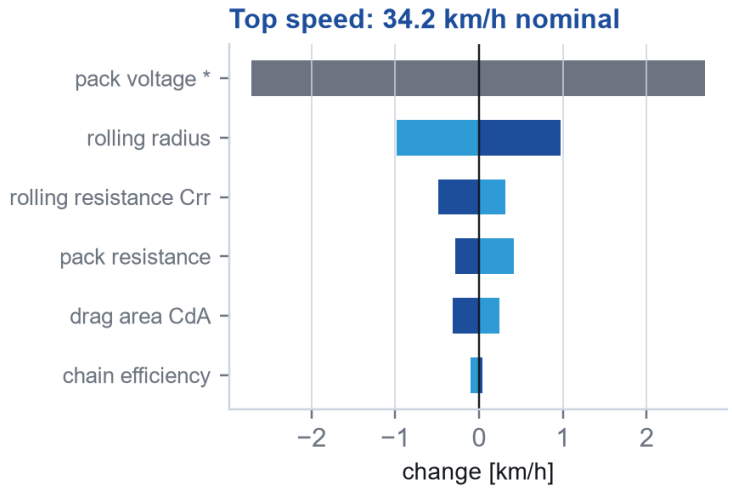


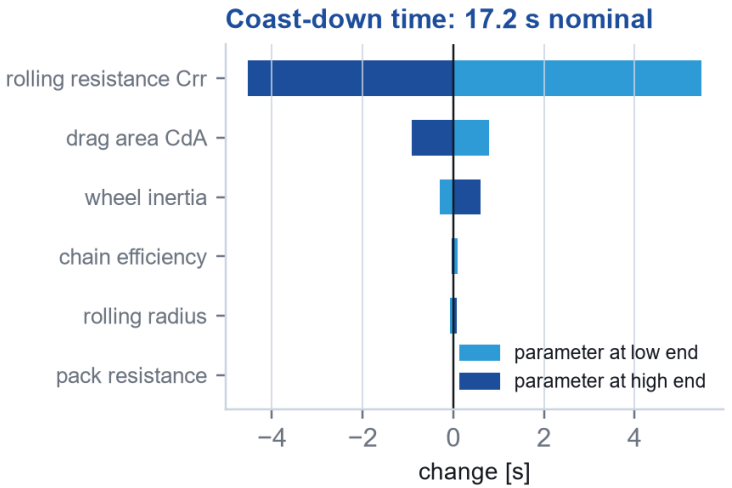




Two simulated manoeuvres. Shaded: every parameter at the ends of its range, one at a time.

# What moves the predictions




**TOP SPEED**

**Pack voltage moves it 31–37 km/h. Measure V_bat first.**

**0 → 25 KM/H**

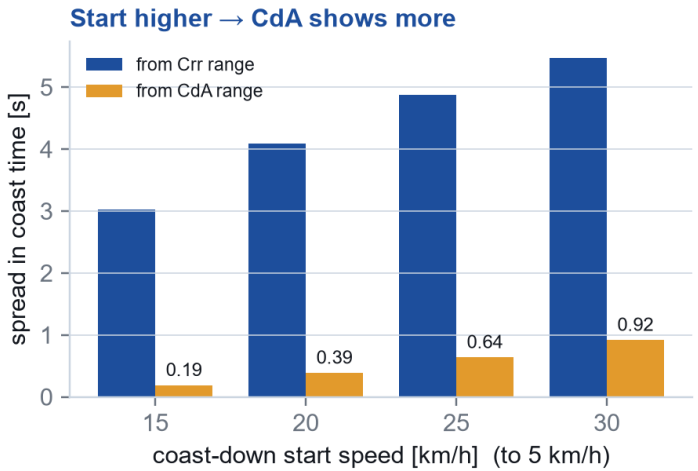
**Grip μ is 60 % of the spread. On dirt the tyre, not the motor.**

**COAST-DOWN**

**Crr is 96 % of the spread. The first thing to measure.**

* pack voltage is a test condition, not an unknown — it is measured, so it stays out of the band.

# Designing the tests




**Start the coast-down as high as the quad reaches: CdA shows 5× more at 30 than at 15 km/h — still small next to Crr.**

**RUN 1 · FULL THROTTLE, 0 → TOP SPEED**

**Checks: top speed · 0 → 25 km/h · battery limit at 9 km/h**

GPS + VESC log · V_bat before

tyres logger

**RUN 2 · COAST-DOWN, 30 → 5 KM/H**

**Checks: coast time. Then fits Crr (CdA only weakly)**

VESC log: speed vs time

tyres logger

**Every run in both directions: slope and wind cancel in the average.**

# What is in the model, and what is next

**Level 1 answers “how fast, how quick”. It cannot answer “how much does it slip”.**

## IN NOW · LEVEL 1

- force balance + grip cap
- 3 limits on iq
- rotor drag
- throttle ramp
- pack R_int sag

## NEXT · TYRES + LEVEL 2

- new tyres → check predictions
- load transfer (needs h_cg)
- launch-control hook
- tyre pressure → Crr

## C02 · MOTOR + INVERTER

- dq machine: Clarke, Park
- current loop
- inverter losses
- temperature

2027 **traction control** **launch control** **MPC**

**For you: 120 A motor and 40 A battery were typed in. How would you derive them?**